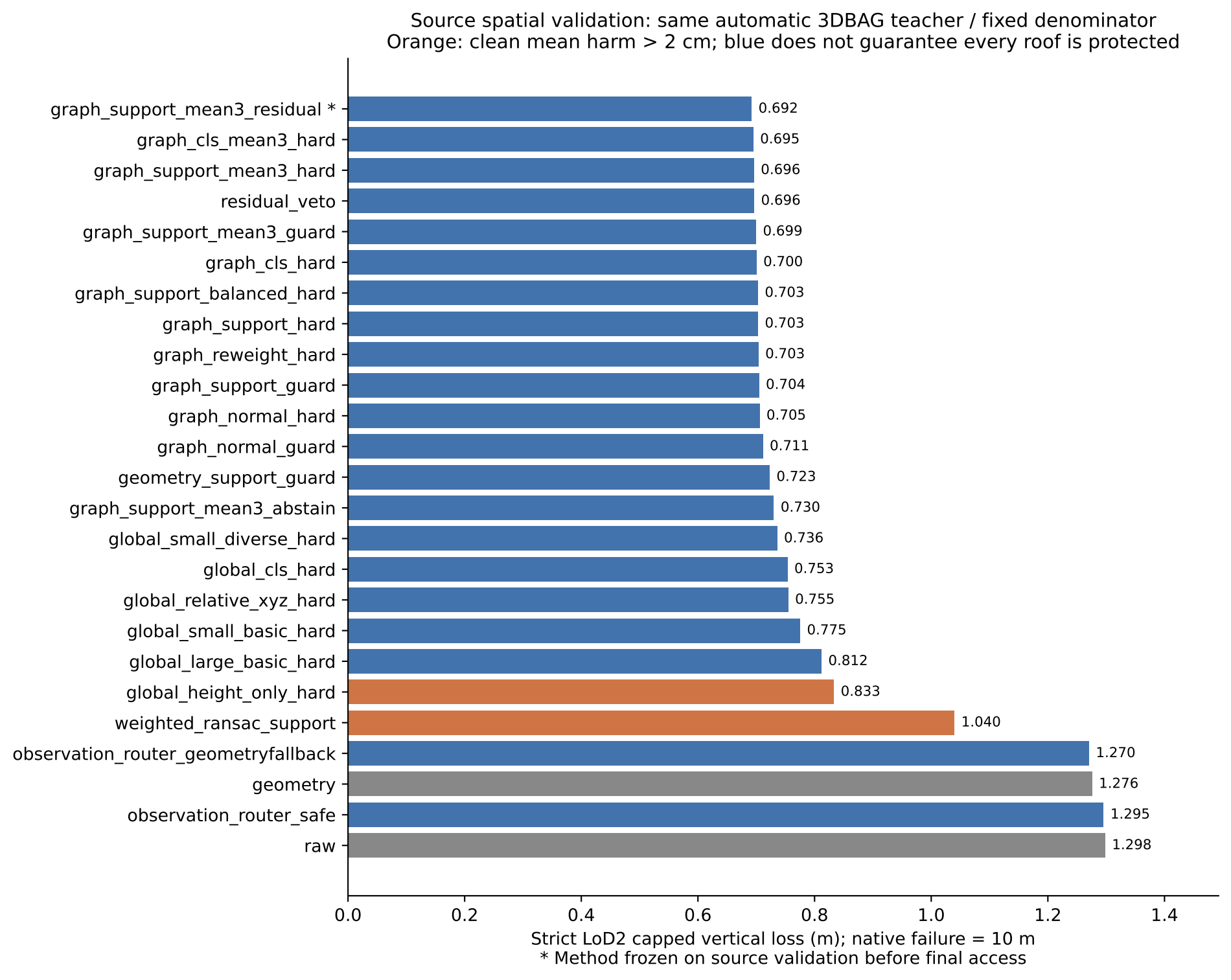
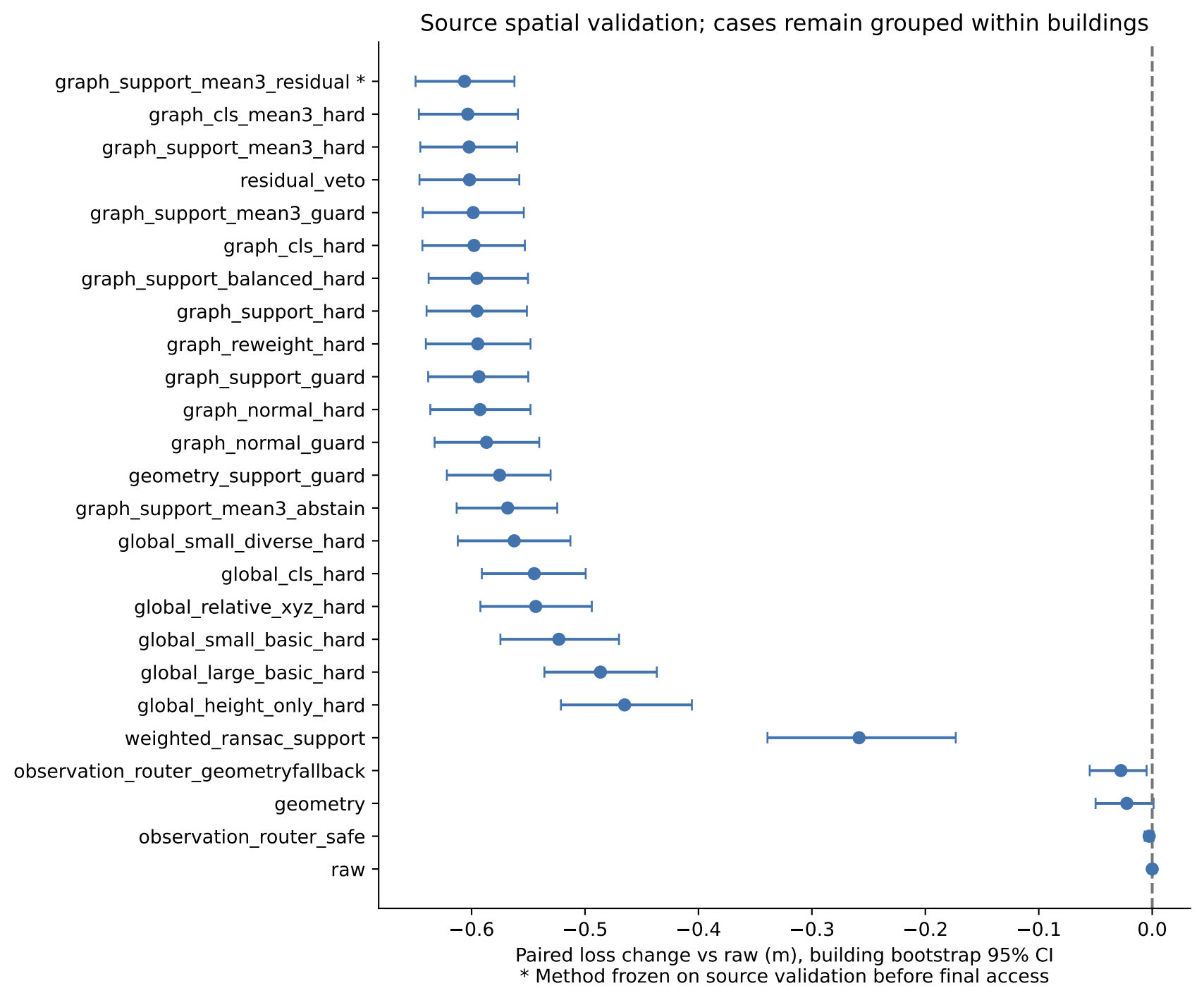
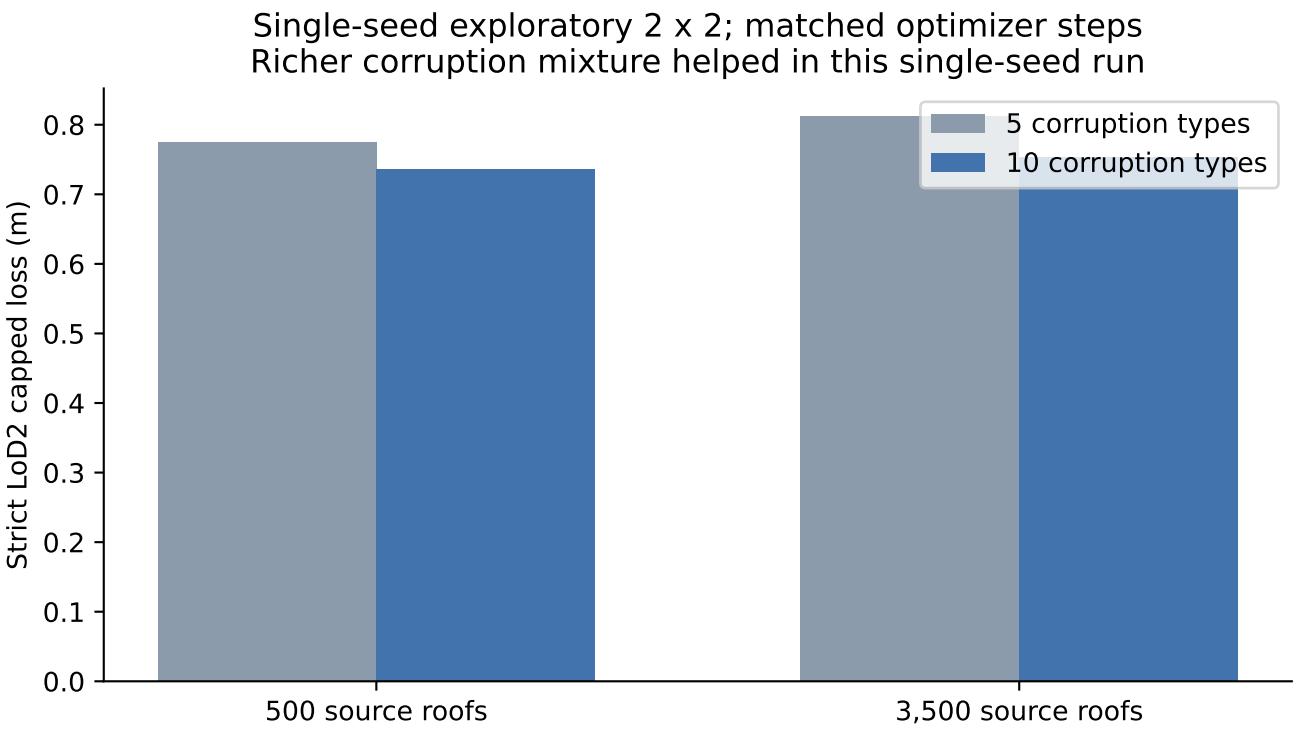
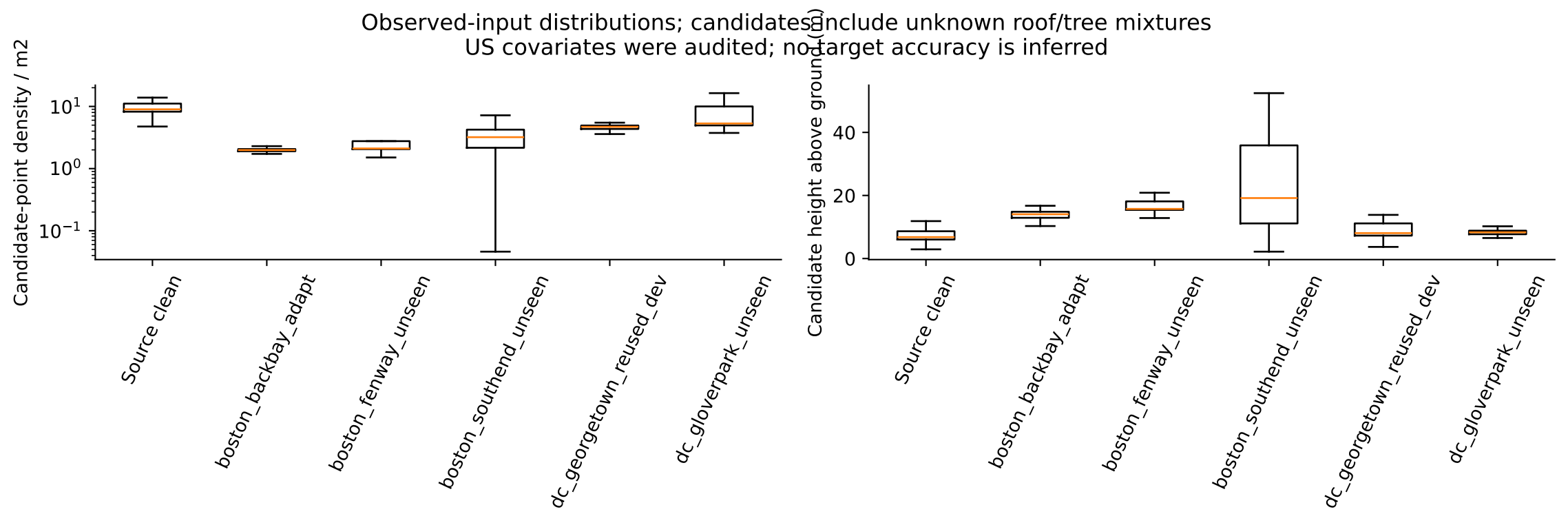
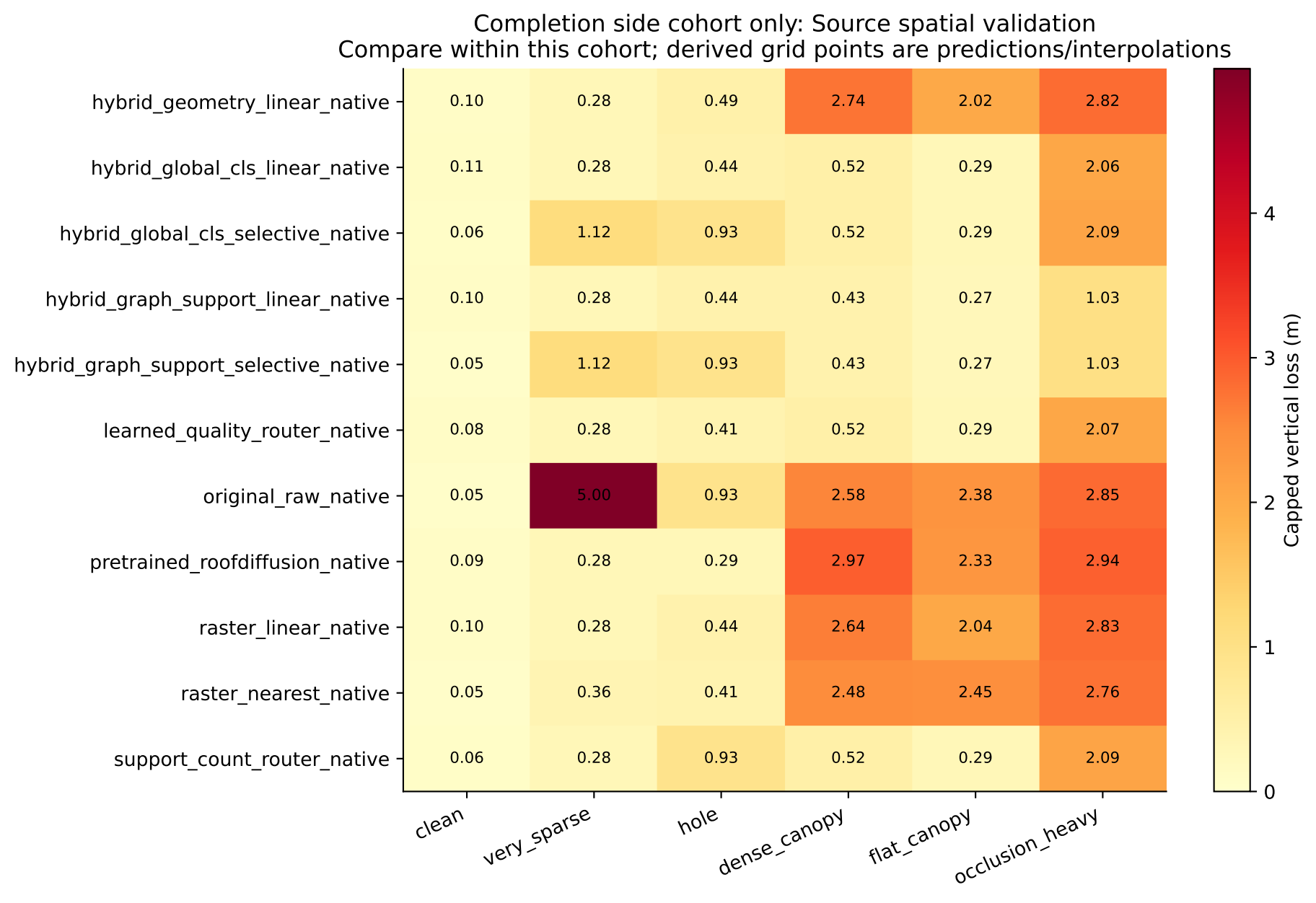

Source spatial validation: same automatic 3DBAG teacher / fixed denominator
Orange: clean mean harm > 2 cm; blue does not guarantee every roof is protected
graph_support_mean3_residual *
graph_cls_mean3_hard
graph_support_mean3_hard
residual_veto
graph_support_mean3_guard
graph_cls_hard
graph_support_balanced_hard
graph_support_hard
graph_reweight_hard
graph_support_guard
graph_normal_hard
graph_normal_guard
geometry_support_guard
graph_support_mean3_abstain
global_small_diverse_hard
global_cls_hard
global_relative_xyz_hard
global_small_basic_hard
global_large_basic_hard
global_height_only_hard
weighted_ransac_support
observation_router_geometryfallback
geometry
observation_router_safe
raw
0.692
0.695
0.696
0.696
0.699
0.700
0.703
0.703
0.703
0.704
0.705
0.711
0.723
0.730
0.736
0.753
0.755
0.775
0.812
0.833
1.040
1.270
1.276
1.295
1.298
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Strict LoD2 capped vertical loss (m); native failure = 10 m
* Method frozen on source validation before final access


Source spatial validation; cases remain grouped within buildings
graph_support_mean3_residual *
graph_cls_mean3_hard
graph_support_mean3_hard
residual_veto
graph_support_mean3_guard
graph_cls_hard
graph_support_balanced_hard
graph_support_hard
graph_reweight_hard
graph_support_guard
graph_normal_hard
graph_normal_guard
geometry_support_guard
graph_support_mean3_abstain
global_small_diverse_hard
global_cls_hard
global_relative_xyz_hard
global_small_basic_hard
global_large_basic_hard
global_height_only_hard
weighted_ransac_support
observation_router_geometryfallback
geometry
observation_router_safe
raw
−0.6
−0.5
−0.4
−0.3
−0.2
−0.1
0.0
Paired loss change vs raw (m), building bootstrap 95% CI
* Method frozen on source validation before final access


Single-seed exploratory 2 x 2; matched optimizer steps
Richer corruption mixture helped in this single-seed run
Strict LoD2 capped loss (m)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
5 corruption types
10 corruption types
500 source roofs
3,500 source roofs


Observed-input distributions; candidates include unknown roof/tree mixtures
US covariates were audited; no target accuracy is inferred
Candidate-point density / m2
10^{1}
10^{0}
10^{-1}
Candidate height above ground (m)
0
20
40
Source clean
boston_backbay_adapt
boston_fenway_unseen
boston_southend_unseen
dc_georgetown_reused_dev
dc_gloverpark_unseen


Completion side cohort only: Source spatial validation
Compare within this cohort; derived grid points are predictions/interpolations
hybrid_geometry_linear_native
hybrid_global_cls_linear_native
hybrid_global_cls_selective_native
hybrid_graph_support_linear_native
hybrid_graph_support_selective_native
learned_quality_router_native
original_raw_native
pretrained_roofdiffusion_native
raster_linear_native
raster_nearest_native
support_count_router_native
0.10 0.28 0.49 2.74 2.02 2.82
0.11 0.28 0.44 0.52 0.29 2.06
0.06 1.12 0.93 0.52 0.29 2.09
0.10 0.28 0.44 0.43 0.27 1.03
0.05 1.12 0.93 0.43 0.27 1.03
0.08 0.28 0.41 0.52 0.29 2.07
0.05 5.00 0.93 2.58 2.38 2.85
0.09 0.28 0.29 2.97 2.33 2.94
0.10 0.28 0.44 2.64 2.04 2.83
0.05 0.36 0.41 2.48 2.45 2.76
0.06 0.28 0.93 0.52 0.29 2.09
clean
very_sparse
hole
dense_canopy
flat_canopy
occlusion_heavy
Capped vertical loss (m)
0
1
2
3
4